

Laboratorium 10.1 — modele efektów mieszanych (**sleepstudy**)

Statystyka w SI i analizie danych — materiał z 8-glmm-1.html

Wojciech Maćkowiak, 412430

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Pakiety: lme4 (lmer, sleepstudy), ggplot2, dplyr. **Instalacja:** `install.packages(c("lme4", "ggplot2", "dplyr"))`.

```
library(lme4)
library(ggplot2)
library(dplyr)
```

Przykład z wykładu — zbiór **sleepstudy**

Dane: czas reakcji w zadaniu psychofizycznym w funkcji dni bez snu (18 osób, 10 pomiarów).

```
head(sleepstudy, 10)
```

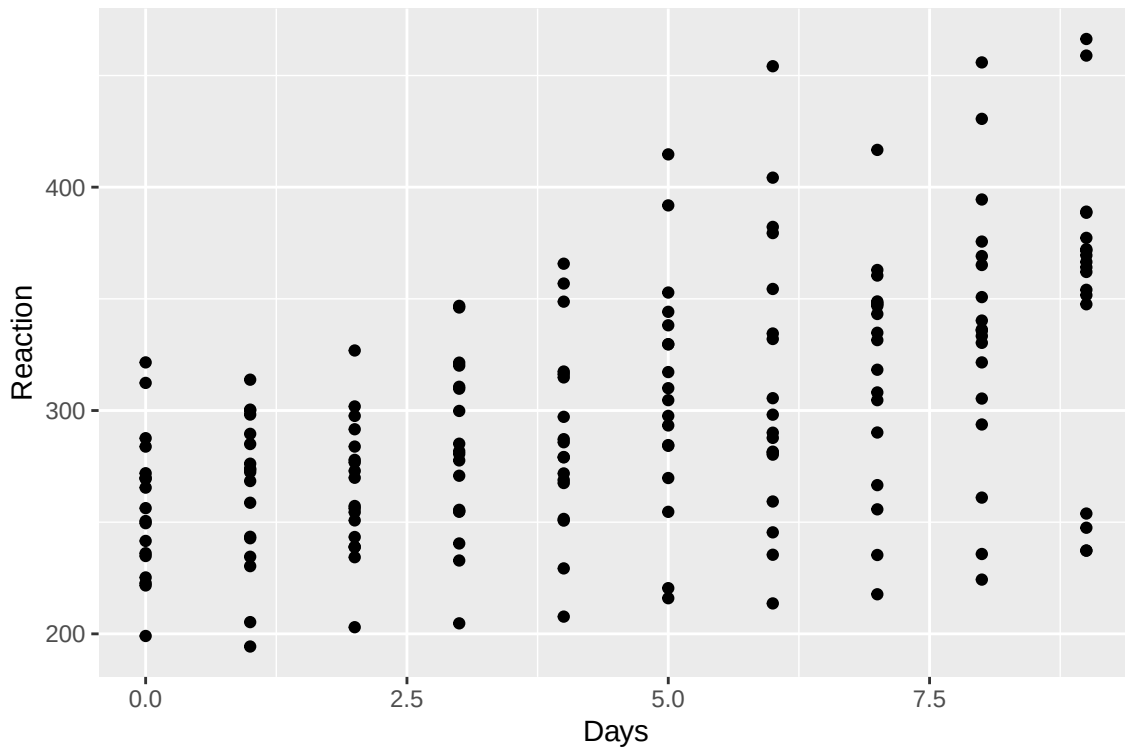
```
##      Reaction Days Subject
## 1  249.5600    0     308
## 2  258.7047    1     308
## 3  250.8006    2     308
## 4  321.4398    3     308
## 5  356.8519    4     308
## 6  414.6901    5     308
## 7  382.2038    6     308
## 8  290.1486    7     308
```

```
## 9 430.5853 8 308
## 10 466.3535 9 308
```

```
tail(sleepstudy, 10)
```

```
##      Reaction Days Subject
## 171 269.4117 0 372
## 172 273.4740 1 372
## 173 297.5968 2 372
## 174 310.6316 3 372
## 175 287.1726 4 372
## 176 329.6076 5 372
## 177 334.4818 6 372
## 178 343.2199 7 372
## 179 369.1417 8 372
## 180 364.1236 9 372
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction)) +
  geom_point()
```



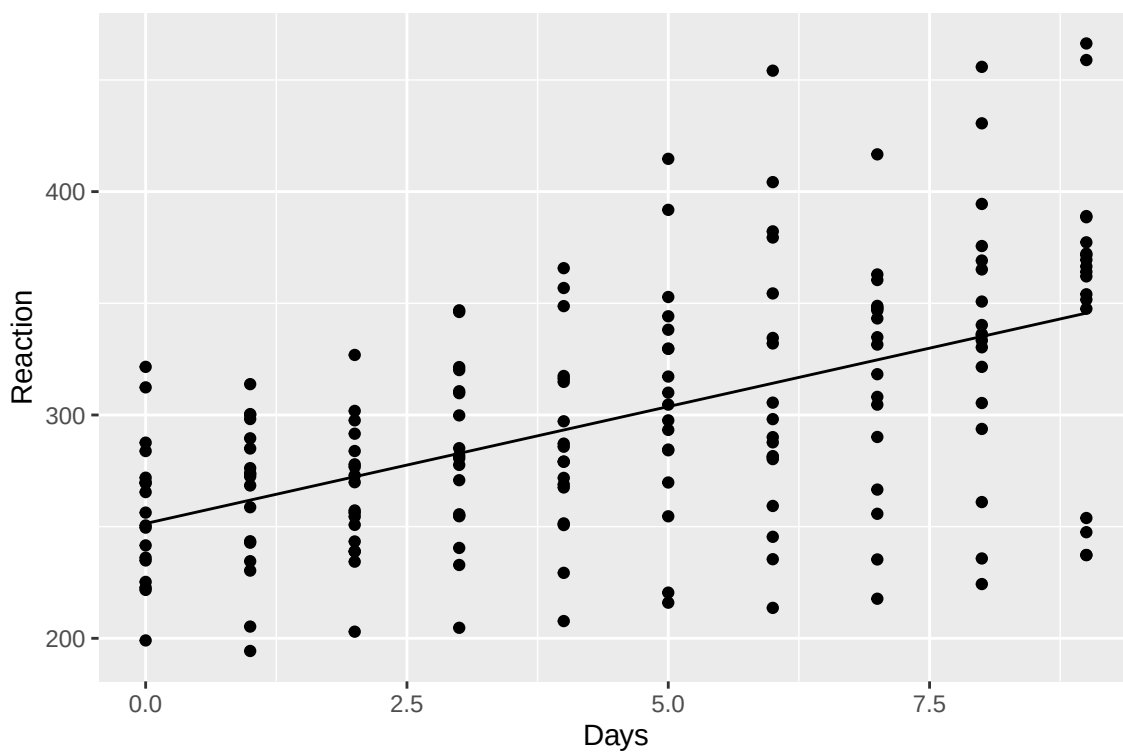
Model liniowy najmniejszych kwadratów

```
ss_lm_fit <- lm(Reaction ~ Days, data = sleepstudy)
summary(ss_lm_fit)
```

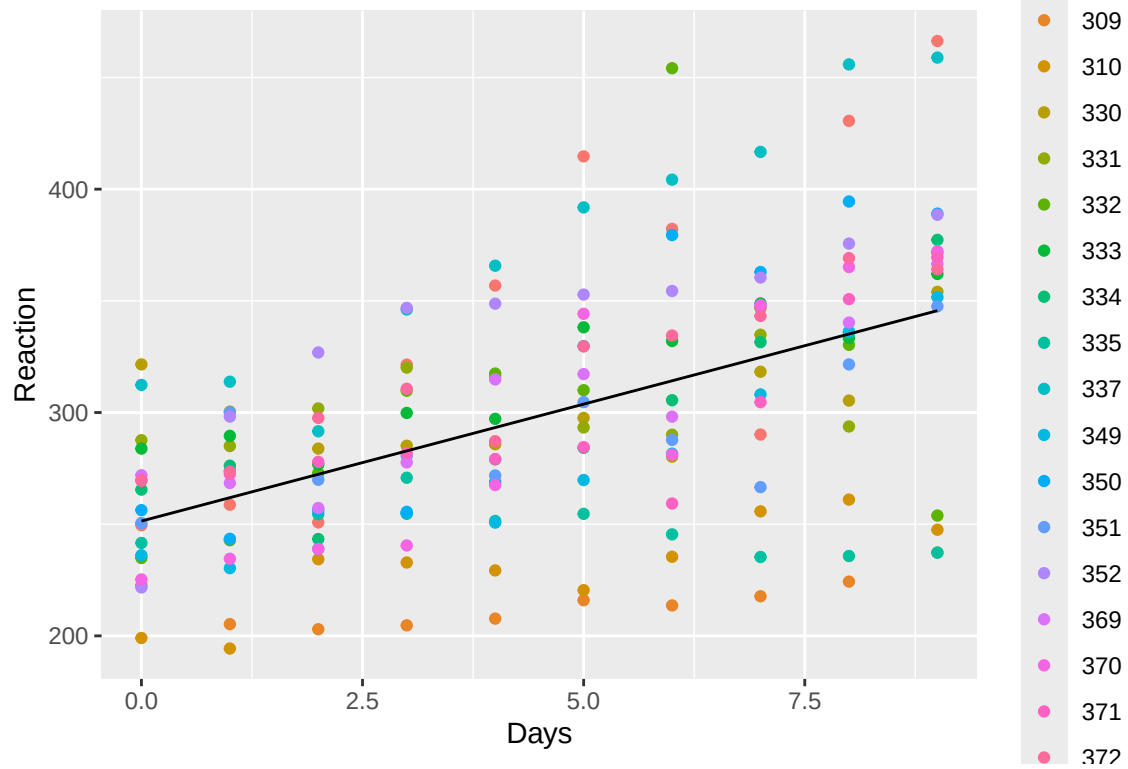
```
##
## Call:
## lm(formula = Reaction ~ Days, data = sleepstudy)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -110.848  -27.483    1.546   26.142  139.953
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   251.405      6.610   38.033 < 2e-16 ***
## Days          10.467      1.238    8.454 9.89e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 47.71 on 178 degrees of freedom
## Multiple R-squared:  0.2865, Adjusted R-squared:  0.2825
## F-statistic: 71.46 on 1 and 178 DF,  p-value: 9.894e-15
```

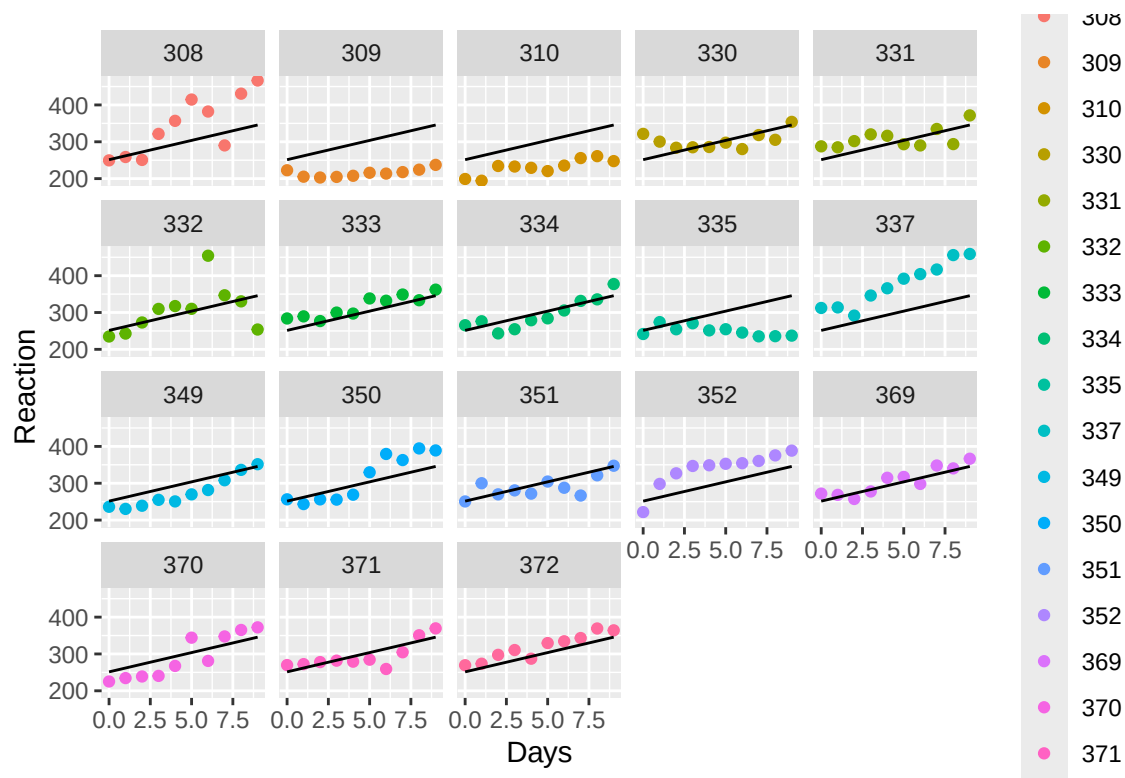
```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction)) +
  geom_point() +
  geom_line(aes(y = predict(ss_lm_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction)) +
  geom_point(aes(color = Subject)) +
  geom_line(aes(y = predict(ss_lm_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction)) +
  geom_point(aes(color = Subject)) +
  geom_line(aes(y = predict(ss_lm_fit))) +
  facet_wrap(~Subject)
```



Model losowych wyrazów wolnych (RI)

```
ss_ri_fit <- lmer(Reaction ~ 1 + (1 | Subject), data = sleepstudy)
summary(ss_ri_fit)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Reaction ~ 1 + (1 | Subject)
## Data: sleepstudy
##
## REML criterion at convergence: 1904.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.4983 -0.5501 -0.1476  0.5123  3.3446
##
## Random effects:
##  Groups   Name      Variance Std.Dev.
## Subject (Intercept) 1278     35.75
## Residual              1959     44.26
## Number of obs: 180, groups: Subject, 18
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   298.51      9.05    32.98
```

```
ss_ri_vcov <- as.data.frame(VarCorr(ss_ri_fit))$vcov
names(ss_ri_vcov) <- c("sigma^2_u", "sigma^2")
ss_ri_vcov
```

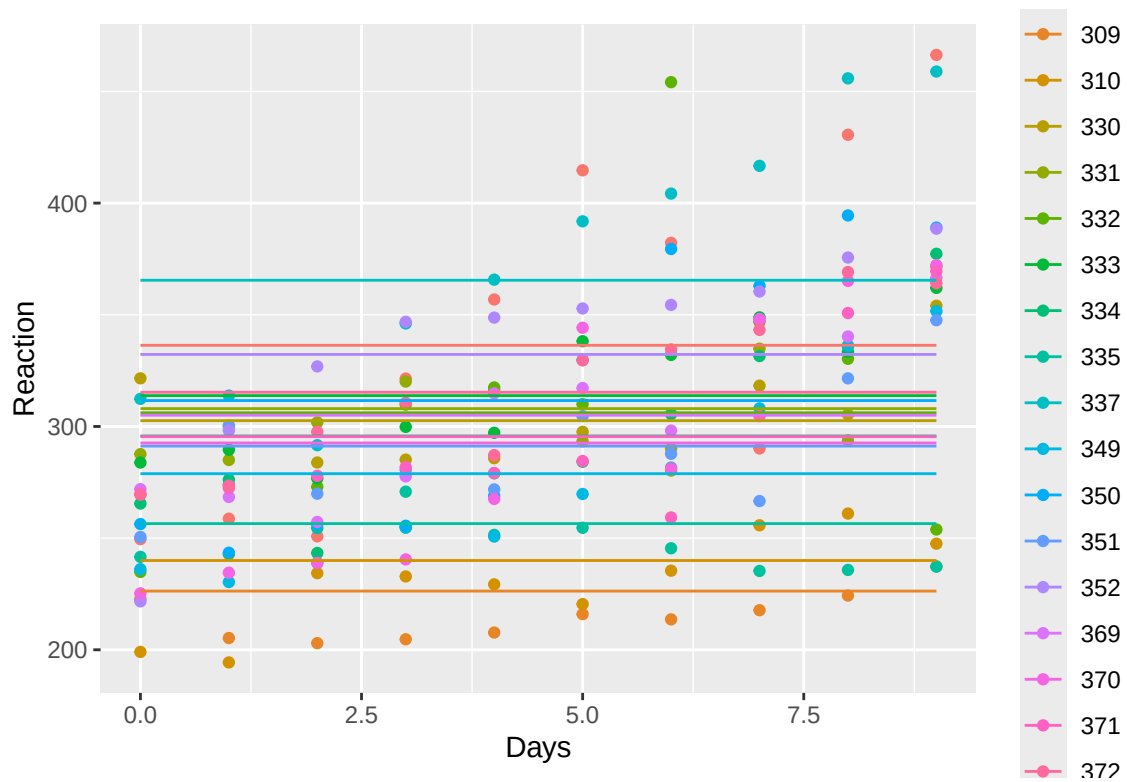
```
## sigma^2_u    sigma^2
## 1278.338    1958.865
```

Udział wariancji między osobami (ICC):

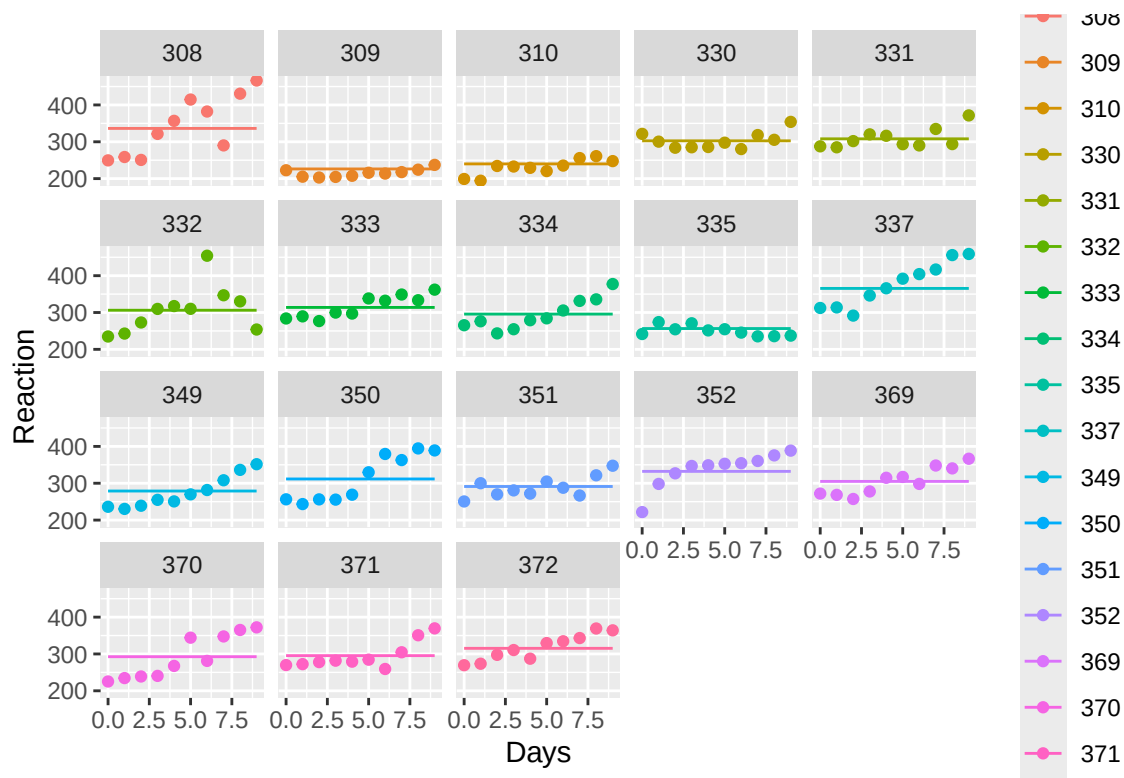
```
ss_ri_vcov["sigma^2_u"] / sum(ss_ri_vcov)
```

```
## sigma^2_u
## 0.3948896
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_ri_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_ri_fit))) +
  facet_wrap(~Subject)
```



Model losowych wyrazów wolnych i współczynników kierunkowych (RSI)

```
ss_rsi_fit <- lmer(Reaction ~ Days + (Days | Subject), data = sleepstudy)
summary(ss_rsi_fit)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Reaction ~ Days + (Days | Subject)
## Data: sleepstudy
##
## REML criterion at convergence: 1743.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9536 -0.4634  0.0231  0.4634  5.1793
##
## Random effects:
## Groups   Name                Variance Std.Dev. Corr
## Subject  (Intercept)        612.10   24.741
##          Days                35.07    5.922   0.07
## Residual                    654.94   25.592
## Number of obs: 180, groups: Subject, 18
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)  251.405     6.825   36.838
## Days         10.467     1.546    6.771
```

```
##
## Correlation of Fixed Effects:
##      (Intr)
## Days -0.138
```

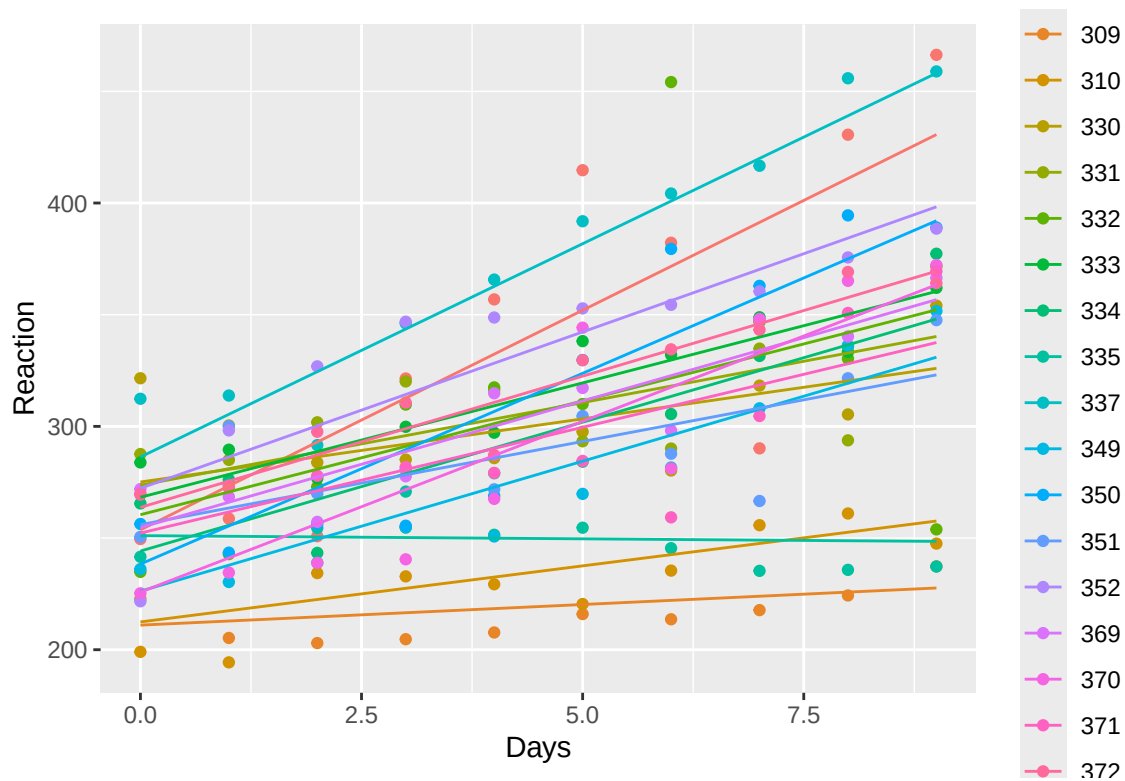
```
fixef(ss_rsi_fit)
```

```
## (Intercept)      Days
##  251.40510    10.46729
```

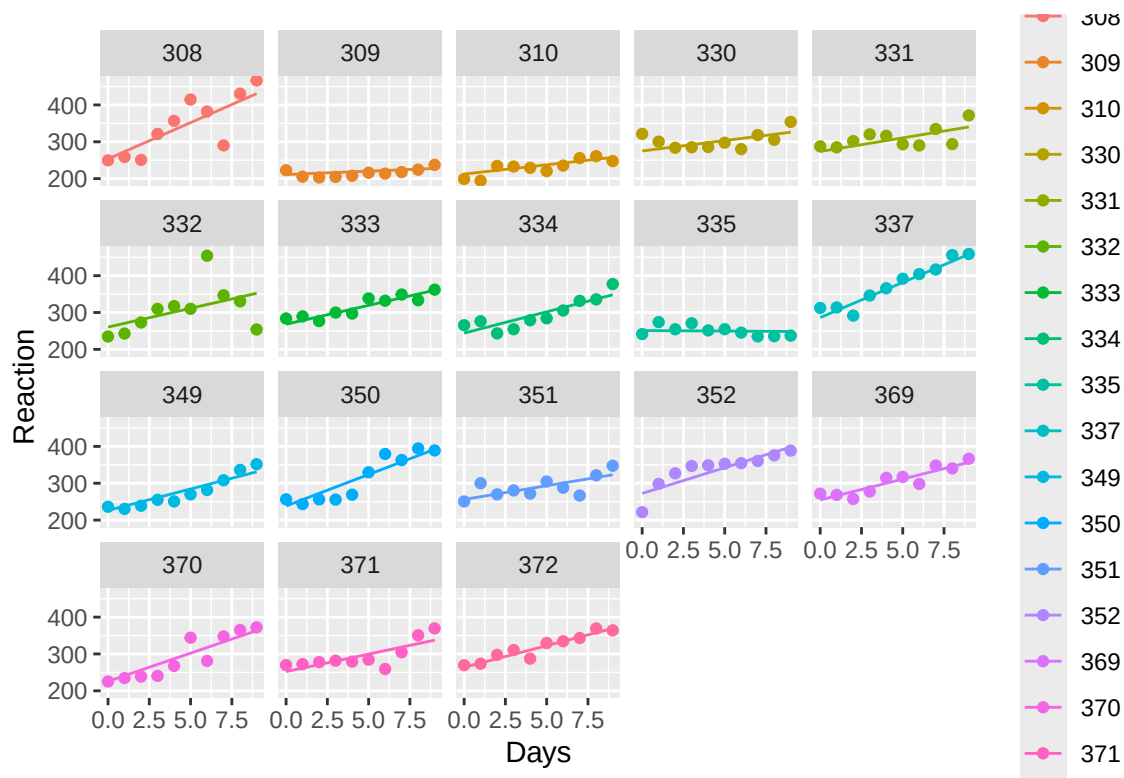
```
ss_rsi_vc <- as.data.frame(VarCorr(ss_rsi_fit))
sigma_rho <- ss_rsi_vc$vcov
sigma_rho[3] <- ss_rsi_vc$sdcor[3]
names(sigma_rho) <- c("sigma^2_u", "sigma^2_v", "rho_uv", "sigma^2")
sigma_rho
```

```
##      sigma^2_u      sigma^2_v      rho_uv      sigma^2
## 612.10015802  35.07171445  0.06555124 654.94000826
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rsi_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rsi_fit))) +
  facet_wrap(~Subject)
```

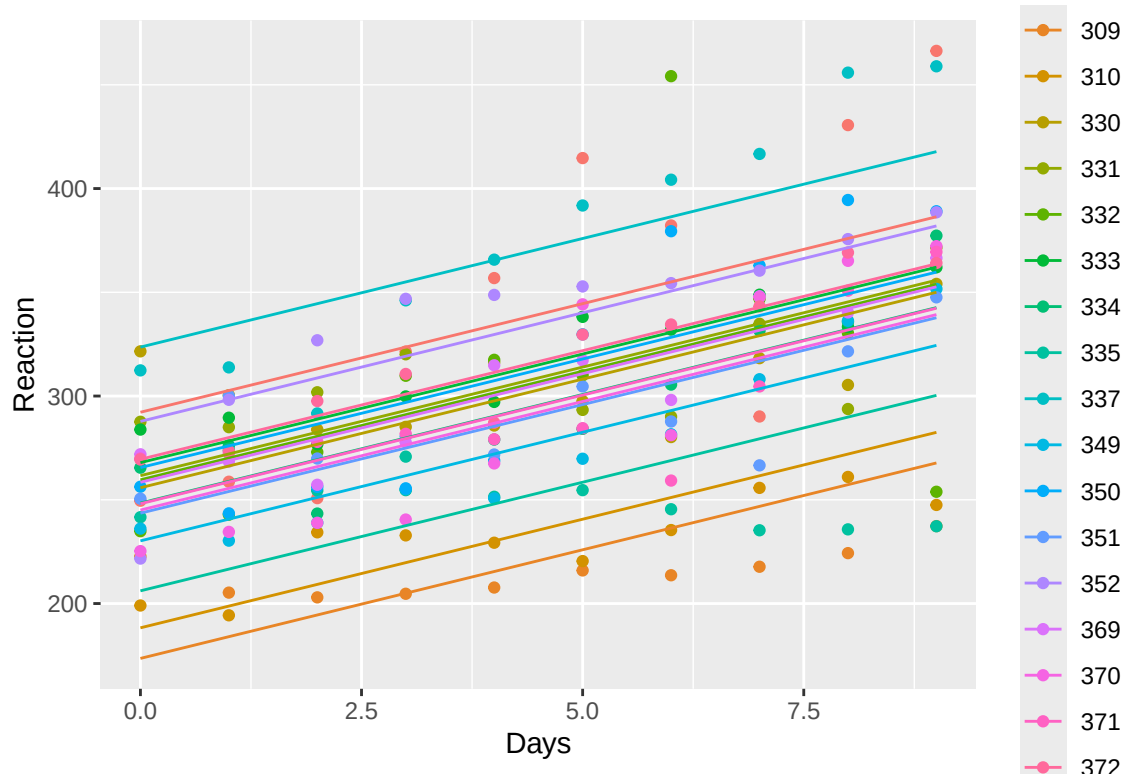
Model pośredni RIFS — ustalony współczynnik kierunkowy, losowe wyrazy wolne

```
ss_rifs_fit <- lmer(Reaction ~ Days + (1 | Subject), data = sleepstudy)
summary(ss_rifs_fit)
```

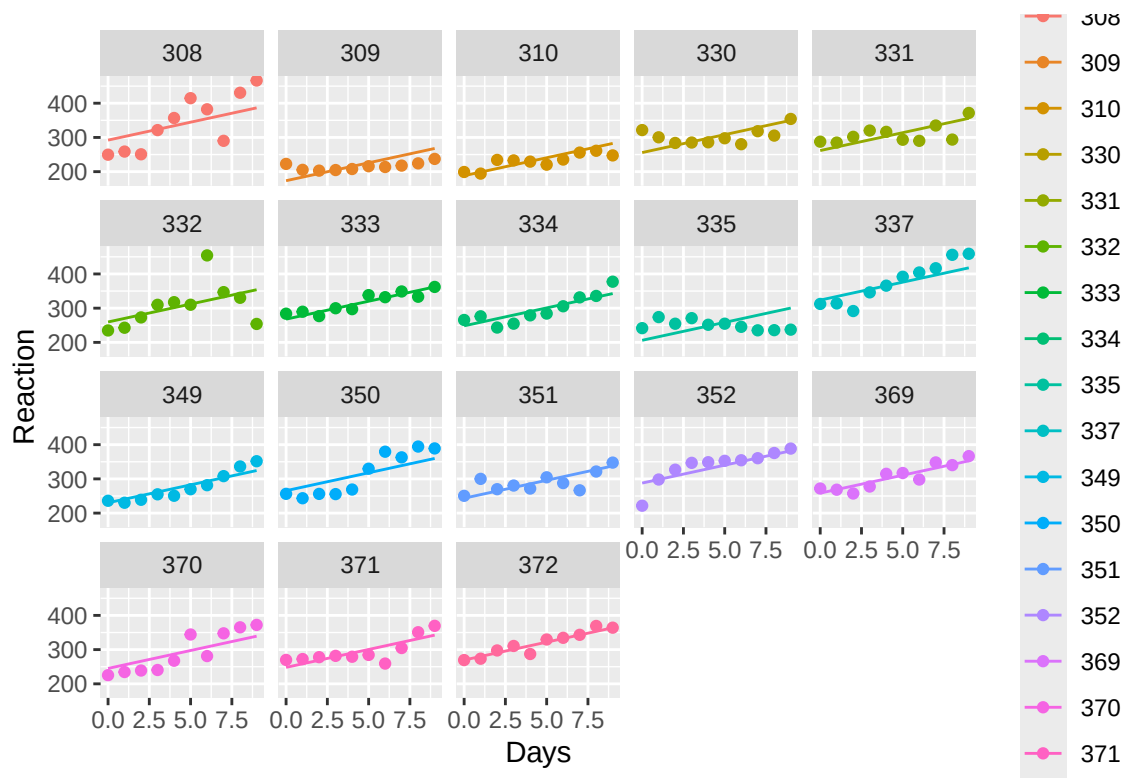
```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Reaction ~ Days + (1 | Subject)
## Data: sleepstudy
##
## REML criterion at convergence: 1786.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2257 -0.5529  0.0109  0.5188  4.2506
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## Subject  (Intercept) 1378.2    37.12
## Residual                    960.5    30.99
## Number of obs: 180, groups: Subject, 18
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept) 251.4051    9.7467   25.79
## Days        10.4673    0.8042   13.02
##
```

```
## Correlation of Fixed Effects:
##      (Intr)
## Days -0.371
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rifs_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rifs_fit))) +
  facet_wrap(~Subject)
```



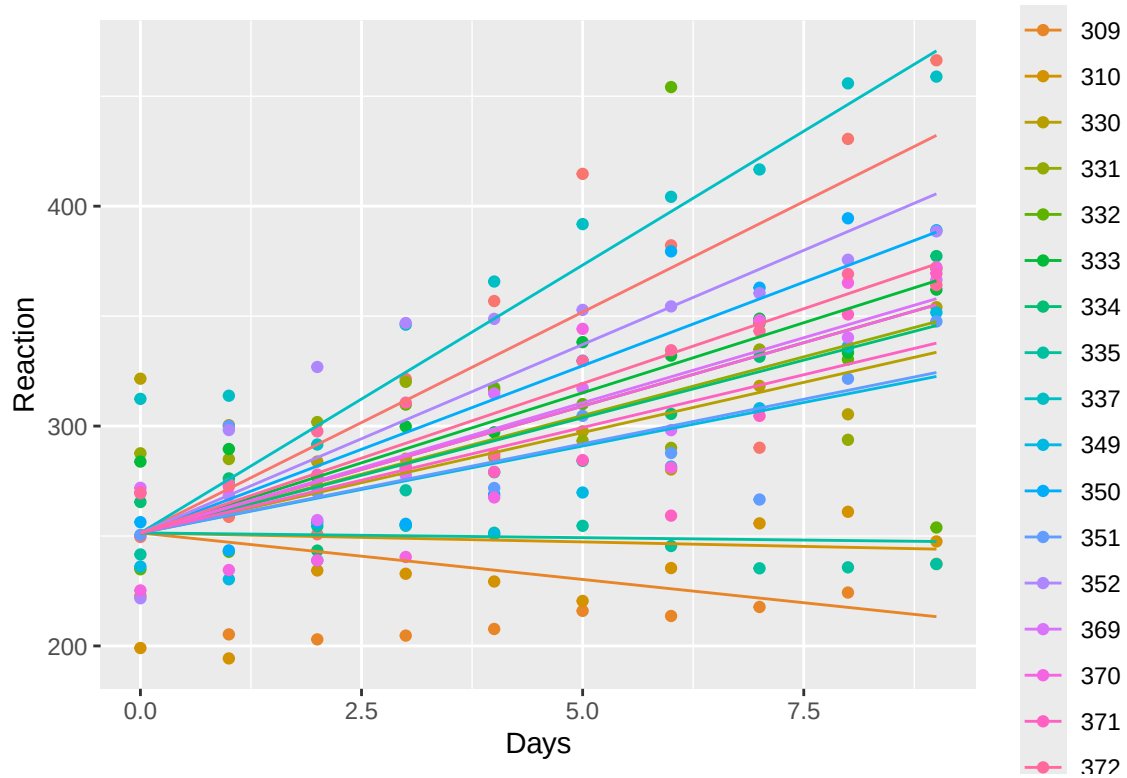
Model pośredni FIRS — losowe współczynniki kierunkowe, ustalony wyraz wolny

```
ss_firs_fit <- lmer(Reaction ~ Days + (0 + Days | Subject), data = sleepstudy)
summary(ss_firs_fit)
```

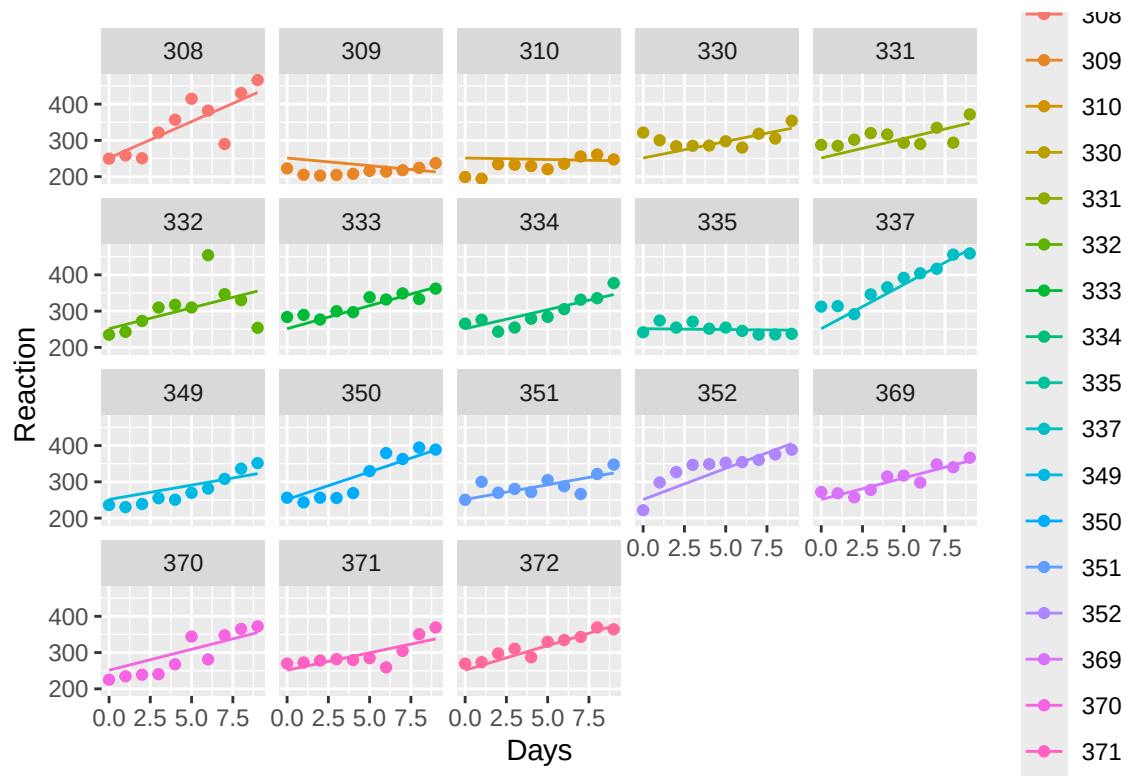
```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Reaction ~ Days + (0 + Days | Subject)
## Data: sleepstudy
##
## REML criterion at convergence: 1766.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5104 -0.5588  0.0541  0.6244  4.6022
##
## Random effects:
##  Groups   Name  Variance Std.Dev.
##  Subject  Days   52.71    7.26
##  Residual             842.03   29.02
## Number of obs: 180, groups:  Subject, 18
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   251.41      4.02   62.539
## Days           10.47      1.87    5.599
##
```

```
## Correlation of Fixed Effects:
##      (Intr)
## Days -0.340
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_firs_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_firs_fit))) +
  facet_wrap(~Subject)
```



Model RSI z niezależnymi błędami na poziomie osoby (RSI-i)

```
ss_rsii_fit <- lmer(Reaction ~ Days + (Days || Subject), data = sleepstudy)
summary(ss_rsii_fit)
```

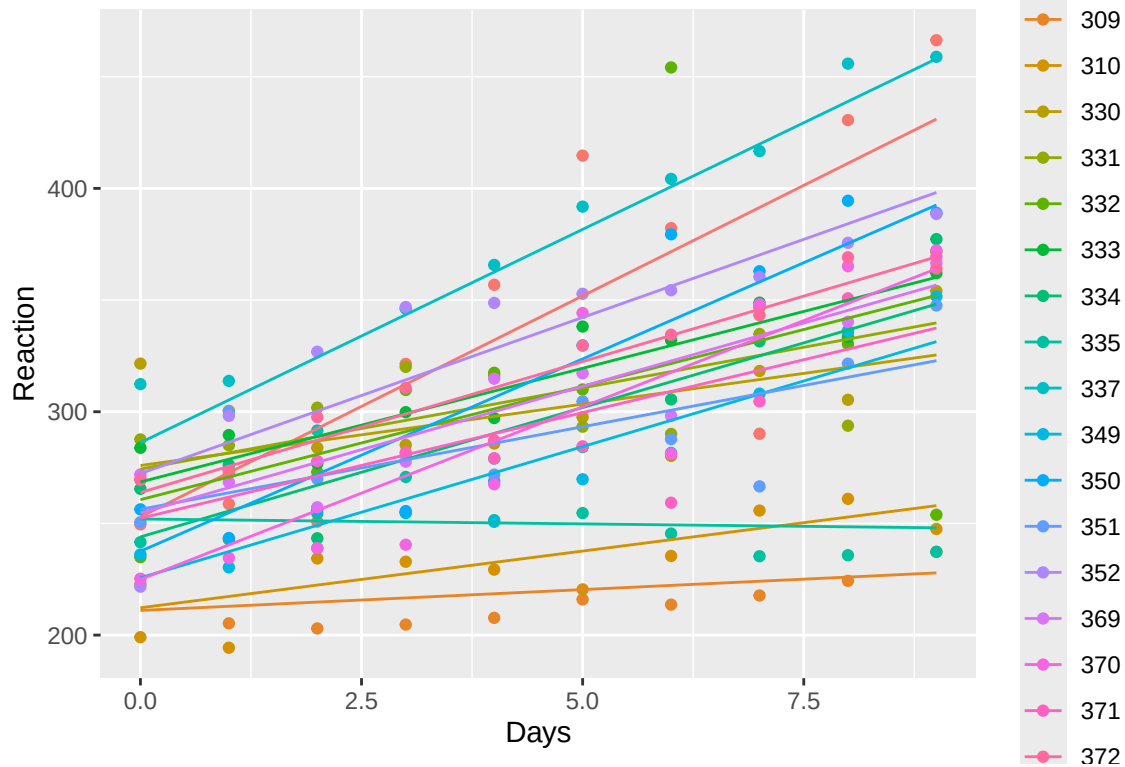
```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Reaction ~ Days + ((1 | Subject) + (0 + Days | Subject))
## Data: sleepstudy
##
## REML criterion at convergence: 1743.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9626 -0.4625  0.0204  0.4653  5.1860
##
## Random effects:
## Groups   Name                Variance Std.Dev.
## Subject  (Intercept)         627.57   25.051
## Subject.1 Days                35.86    5.988
## Residual                    653.58   25.565
## Number of obs: 180, groups: Subject, 18
##
## Fixed effects:
##              Estimate Std. Error t value
## (Intercept)   251.405      6.885   36.513
## Days           10.467      1.560    6.712
##
## Correlation of Fixed Effects:
```

```
##      (Intr)
## Days -0.184

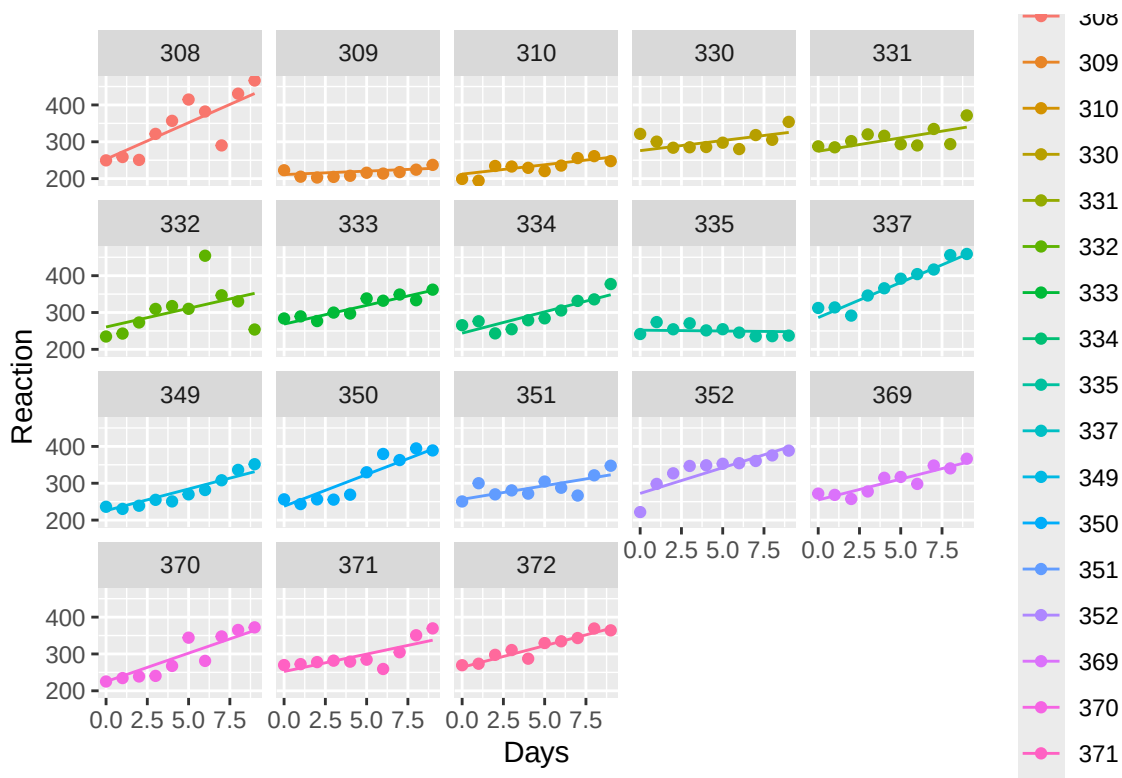
ss_rsii_vcov <- as.data.frame(VarCorr(ss_rsii_fit))$vcov
names(ss_rsii_vcov) <- c("sigma^2_u", "sigma^2_v", "sigma^2")
ss_rsii_vcov
```

```
## sigma^2_u sigma^2_v  sigma^2
## 627.56905  35.85838 653.58350
```

```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rsii_fit)))
```



```
sleepstudy |>
  ggplot(aes(x = Days, y = Reaction, color = Subject)) +
  geom_point() +
  geom_line(aes(y = predict(ss_rsii_fit))) +
  facet_wrap(~Subject)
```



Porównanie modeli

```
anova(ss_ri_fit, ss_rsii_fit, ss_rifs_fit, ss_firs_fit, ss_ri_fit)
```

```
## Data: sleepstudy
## Models:
## ss_ri_fit: Reaction ~ 1 + (1 | Subject)
## ss_rifs_fit: Reaction ~ Days + (1 | Subject)
## ss_firs_fit: Reaction ~ Days + (0 + Days | Subject)
## ss_rsii_fit: Reaction ~ Days + ((1 | Subject) + (0 + Days | Subject))
## ss_rsi_fit: Reaction ~ Days + (Days | Subject)
##
```

	npars	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
## ss_ri_fit	3	1916.5	1926.1	-955.27	1910.5			
## ss_rifs_fit	4	1802.1	1814.8	-897.04	1794.1	116.4624	1	< 2.2e-16 ***
## ss_firs_fit	4	1782.1	1794.8	-887.04	1774.1	19.9983	0	
## ss_rsii_fit	5	1762.0	1778.0	-876.00	1752.0	22.0771	1	2.619e-06 ***
## ss_rsi_fit	6	1763.9	1783.1	-875.97	1751.9	0.0639	1	0.8004

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Powyższa tabela `anova` z pakietu `lme4` porównuje modele zagnieżdżone pod względem losowej struktury (testy odchylenia). W razie potrzeby można uzupełnić porównanie kryteriami AIC/BIC (`AIC()`, `BIC()`).